# ConstructPro

## Test Case Register

Requirements traced to automated tests
SENG 34213 - System Development Project

By

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# Chapter 1 - Overview

This register maps the selected SRS requirements to the automated tests for ConstructPro ERP, following the test-case fields required by Section 6.3.3 of the SENG 34213 guideline. It retains the 23 cases from the original Test Case Register and Appendix A of the Final Development Report.

The backend Unit and Integration case results are retained from the development report's recorded CI evidence (backend develop commit f6bf853). The Playwright case results are updated from the local execution recorded on 25 October 2026. The current GitHub Actions check performs TypeScript validation and Playwright discovery; it does not establish a passing deployed E2E suite.

Table 1.1: Test cases by type and status

| Type | Cases | Pass | Fail | Skipped |
|---|---|---|---|---|
| Unit | 8 | 8 | 0 | 0 |
| Integration | 5 | 5 | 0 | 0 |
| E2E (API / UI / Security) | 10 | 4 | 4 | 2 |
| Total | 23 | 17 | 4 | 2 |

## Execution results and evidence

The local Playwright execution covered 107 tests in 23 specification files: 83 passed, 7 failed, 16 were skipped and 1 did not run. Those suite-wide figures should not be confused with the 23 selected, requirement-traced cases in Table 1.1.

The seven failures were concentrated in AI Analytics, two Finance/AI authorization checks, the finance payment-to-dashboard workflow, and three Quotation frontend scenarios (FE-02, FE-03 and FE-05). The register includes four of these failures; FE-03, FE-05 and the separate anonymous-authorization case are outside its 23 selected entries.

The attached test-results.zip contains execution failure traces, screenshots and videos. Its embedded Playwright HTML report reflects a subsequent discovery/listing operation rather than the recorded full execution, so it is not used as independent proof of the overall pass count.

Table 1.2: Test case summary

| Test Case ID | Test Case Name | Type | Status |
|---|---|---|---|
| TC-FR-001-01 | Login issues access and refresh tokens | Unit | Pass |
| TC-FR-001-02 | Create user, assign role, no password in response | E2E | Pass |
| TC-FR-001-03 | Reject duplicate user email | Unit | Pass |
| TC-FR-002-01 | Capture, search, edit and delete a lead | E2E | Pass |
| TC-FR-003-01 | Quotation total calculated on the server | E2E (API) | Pass |
| TC-FR-003-02 | Live total updates while adding items | E2E (UI) | Fail |
| TC-FR-003-03 | Rejection requires a reason | E2E (UI) | Pass |
| TC-FR-004-01 | Duplicate conversion is blocked | E2E (API) | Skipped |
| TC-FR-004-02 | Concurrent conversions create one project | Integration | Pass |
| TC-FR-005-01 | Weighted progress uses relative weights | Unit | Pass |
| TC-FR-005-02 | Invalid milestone weight rejected | Unit | Pass |
| TC-FR-005-03 | Project cannot complete with incomplete milestones | Unit | Pass |
| TC-FR-006-01 | Duplicate and excessive payments rolled back | Integration | Pass |
| TC-FR-006-02 | Payment updates invoice and dashboard KPIs | E2E | Fail |
| TC-FR-007-01 | Revenue KPIs from invoice aggregates | Unit | Pass |
| TC-FR-008-01 | Oversized document upload rejected | Integration | Pass |
| TC-FR-008-02 | Unauthenticated document request rejected | Integration | Pass |
| TC-FR-009-01 | Calendar refresh token stored encrypted | Unit | Pass |
| TC-FR-010-01 | AI prediction falls back safely | Unit | Pass |
| TC-FR-010-02 | AI prediction shown in Analytics | E2E | Fail |
| TC-NFR-08-01 | Client portal hides other customers' data | Integration | Pass |
| TC-NFR-08-02 | Non-finance role cannot use finance APIs | E2E (Security) | Fail |
| TC-NFR-08-03 | Accountant cannot list users or leads | E2E (Security) | Skipped |

# Chapter 2 - Test Cases

Each case retains the eleven fields from the original register. The Actual Output and Status fields have been reconciled with the available test evidence; skipped or failed cases are not marked as passed.

| Field | Content |
|---|---|
| Test Case ID | TC-FR-001-01 |
| Test Case Name | Login issues access and refresh tokens |
| Related Requirement | FR-001 (Manage Users), NFR-07 (JWT Authentication) |
| Related Test File | `backend/test/auth-service/google-session.spec.ts` |
| Type | Unit |
| Priority | P1 – Must Pass |
| Preconditions | Active user exists |
| Input | Valid sign-in |
| Expected Output | Access token valid 15 min and a separate refresh token valid 7 days |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-001-02 |
| Test Case Name | Create user, assign role, no password in response |
| Related Requirement | FR-001 (Manage Users), UC-01 |
| Related Test File | `tests/e2e/user-management.spec.ts` |
| Type | E2E |
| Priority | P1 – Must Pass |
| Preconditions | Logged in as ADMIN |
| Input | New user form with role |
| Expected Output | User created with role; response has no password field |
| Actual Output | User creation, role assignment and password-exclusion assertions passed in the local Playwright execution. |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-001-03 |
| Test Case Name | Reject duplicate user email |
| Related Requirement | FR-001 (Manage Users) |
| Related Test File | `backend/test/user-service/user.service.spec.ts` |
| Type | Unit |
| Priority | P2 – Should Pass |
| Preconditions | User with the email exists |
| Input | Create user with same email |
| Expected Output | 409 conflict; original account unchanged |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-002-01 |
| Test Case Name | Capture, search, edit and delete a lead |
| Related Requirement | FR-002 (Manage Sales Leads), UC-02 |
| Related Test File | `tests/e2e/leads.spec.ts` |
| Type | E2E |
| Priority | P1 – Must Pass |
| Preconditions | Logged in as SALES_MANAGER or ADMIN |
| Input | Lead details, search term, note |
| Expected Output | Lead persists through each step and is removed on delete |
| Actual Output | Browser-based lead capture, search, editing and deletion workflow passed. |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-003-01 |
| Test Case Name | Quotation total calculated on the server |
| Related Requirement | FR-003 (Generate Quotations), UC-03 |
| Related Test File | `tests/e2e/quotations/quotations-backend.spec.ts (BE-01)` |
| Type | E2E (API) |
| Priority | P1 – Must Pass |
| Preconditions | Lead exists |
| Input | POST /quotations with multiple line items |
| Expected Output | 201; total equals the sum of quantity × unit price |
| Actual Output | BE-01 passed; quotation API accepted the input and validated the line-item total. |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-003-02 |
| Test Case Name | Live total updates while adding items |
| Related Requirement | FR-003 (Generate Quotations) |
| Related Test File | `tests/e2e/quotations/quotations-frontend.spec.ts (FE-02)` |
| Type | E2E (UI) |
| Priority | P2 – Should Pass |
| Preconditions | Quotation form open |
| Input | Add and edit line items |
| Expected Output | Displayed total updates immediately |
| Actual Output | FE-02 timed out while waiting for the "Item 2 unit price" input; live-total assertion was not completed. |
| Status | Fail |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-003-03 |
| Test Case Name | Rejection requires a reason |
| Related Requirement | FR-003 (Generate Quotations), BR 10.2 |
| Related Test File | `tests/e2e/quotations/quotations-frontend.spec.ts (FE-06)` |
| Type | E2E (UI) |
| Priority | P2 – Should Pass |
| Preconditions | Quotation PENDING_APPROVAL |
| Input | Reject without a reason |
| Expected Output | Validation error; status unchanged |
| Actual Output | FE-06 passed; the rejection action remained disabled until a reason was entered. |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-004-01 |
| Test Case Name | Duplicate conversion is blocked |
| Related Requirement | FR-004 (Convert Quotation to Project), BR 10.2 |
| Related Test File | `tests/e2e/quotations/quotations-backend.spec.ts (BE-07)` |
| Type | E2E (API) |
| Priority | P1 – Must Pass |
| Preconditions | Quotation already CONVERTED |
| Input | Approve and convert again |
| Expected Output | 409 ALREADY_CONVERTED with the existing project ID |
| Actual Output | BE-07 skipped: E2E_ALLOW_PERSISTENT_LIFECY-CLE_WRITES was not enabled for persistent conversion writes. |
| Status | Skipped |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-004-02 |
| Test Case Name | Concurrent conversions create one project |
| Related Requirement | FR-004 (Convert Quotation to Project), UC-04 |
| Related Test File | `backend/test/project-service/project- service.integration.spec.ts` |
| Type | Integration |
| Priority | P1 – Must Pass |
| Preconditions | Approved quotation |
| Input | Two simultaneous conversion requests |
| Expected Output | Both return the same project; one project row exists |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-005-01 |
| Test Case Name | Weighted progress uses relative weights |
| Related Requirement | FR-005 (Track Project Milestones), UC-05 |
| Related Test File | `backend/test/project-service/ milestone.service.spec.ts` |
| Type | Unit |
| Priority | P1 – Must Pass |
| Preconditions | Project with milestones of different weights |
| Input | Update milestone progress |
| Expected Output | Project progress = Σ(progress × weight) / Σ weight |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-005-02 |
| Test Case Name | Invalid milestone weight rejected |
| Related Requirement | FR-005 (Track Project Milestones) |
| Related Test File | `backend/test/project-service/ milestone.controller.spec.ts` |
| Type | Unit |
| Priority | P2 – Should Pass |
| Preconditions | Project exists |
| Input | Weight 0 or 11 |
| Expected Output | 400 validation error |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-005-03 |
| Test Case Name | Project cannot complete with incomplete milestones |
| Related Requirement | FR-005 (Track Project Milestones), BR 10.4 |
| Related Test File | `backend/test/project-service/ project.service.spec.ts` |
| Type | Unit |
| Priority | P1 – Must Pass |
| Preconditions | ACTIVE project, one milestone IN_PROGRESS |
| Input | Set status COMPLETED |
| Expected Output | Rejected with a lifecycle error |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-006-01 |
| Test Case Name | Duplicate and excessive payments rolled back |
| Related Requirement | FR-006 (Track Payments), BR 10.3 |
| Related Test File | `backend/test/finance/finance-workflow.db.spec.ts` |
| Type | Integration |
| Priority | P1 – Must Pass |
| Preconditions | ISSUED invoice with outstanding balance |
| Input | Payment above balance; repeated reference |
| Expected Output | 422 / 409; persisted balances unchanged |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-006-02 |
| Test Case Name | Payment updates invoice and dashboard KPIs |
| Related Requirement | FR-006 (Track Payments), FR-007 (KPI Dashboard), UC-06 |
| Related Test File | `tests/e2e/finance-dashboard.spec.ts` |
| Type | E2E |
| Priority | P1 – Must Pass |
| Preconditions | Logged in as ACCOUNTANT; ISSUED invoice |
| Input | Record partial payment |
| Expected Output | Invoice PARTIALLY_PAID, balance reduced, KPIs and UI consistent |
| Actual Output | Finance workflow could not execute: E2E_INVOICE_ID was missing in the recorded test run. |
| Status | Fail |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-007-01 |
| Test Case Name | Revenue KPIs from invoice aggregates |
| Related Requirement | FR-007 (KPI Dashboard), UC-07 |
| Related Test File | `backend/test/analytics-service/ analytics.service.spec.ts` |
| Type | Unit |
| Priority | P2 – Should Pass |
| Preconditions | Invoices and payments exist |
| Input | GET revenue KPIs |
| Expected Output | Totals match invoice aggregates |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-008-01 |
| Test Case Name | Oversized document upload rejected |
| Related Requirement | FR-008 (Store Project Documents), UC-08 |
| Related Test File | `backend/test/document-service/document- gateway.e2e-spec.ts` |
| Type | Integration |
| Priority | P2 – Should Pass |
| Preconditions | Logged in as PROJECT_MANAGER |
| Input | File larger than the limit |
| Expected Output | 413 with a document-specific error code |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-008-02 |
| Test Case Name | Unauthenticated document request rejected |
| Related Requirement | FR-008 (Store Project Documents), NFR-08 (Role-Based Access Control) |
| Related Test File | `backend/test/document-service/document- gateway.e2e-spec.ts` |
| Type | Integration |
| Priority | P1 – Must Pass |
| Preconditions | None |
| Input | Request without token |
| Expected Output | 401 TOKEN_MISSING |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-009-01 |
| Test Case Name | Calendar refresh token stored encrypted |
| Related Requirement | FR-009 (Google Calendar Integration), UC-09 |
| Related Test File | `backend/test/calender-service/ calendar.service.spec.ts` |
| Type | Unit |
| Priority | P1 – Must Pass |
| Preconditions | User completes Google consent |
| Input | Store connection |
| Expected Output | Token stored as AES-256-GCM ciphertext with IV and tag |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-010-01 |
| Test Case Name | AI prediction falls back safely |
| Related Requirement | FR-010 (AI Risk Prediction), UC-10 |
| Related Test File | `backend/test/ai-service/ai- forecasting.service.spec.ts` |
| Type | Unit |
| Priority | P2 – Should Pass |
| Preconditions | Project with little history |
| Input | Request risk prediction |
| Expected Output | SAFE_FALLBACK result with explanation and low confidence |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-FR-010-02 |
| Test Case Name | AI prediction shown in Analytics |
| Related Requirement | FR-010 (AI Risk Prediction), UC-10 |
| Related Test File | `tests/e2e/ai-analytics.spec.ts` |
| Type | E2E |
| Priority | P2 – Should Pass |
| Preconditions | Logged in as management user |
| Input | Generate prediction |
| Expected Output | Contract-valid prediction displayed |
| Actual Output | AI prediction request proceeded, but the UI check failed due to an ambiguous "Revenue trend" text locator. |
| Status | Fail |

| Field | Content |
|---|---|
| Test Case ID | TC-NFR-08-01 |
| Test Case Name | Client portal hides other customers' data |
| Related Requirement | NFR-08 (Role-Based Access Control), ADR-15 |
| Related Test File | `backend/test/client-portal/client-portal- owner-ship.integration.spec.ts` |
| Type | Integration |
| Priority | P1 – Must Pass |
| Preconditions | Two customers with projects |
| Input | Portal user lists projects and documents |
| Expected Output | Only own projects returned; mixed-customer projects hidden |
| Actual Output | As expected (backend CI evidence in Final Development Report). |
| Status | Pass |

| Field | Content |
|---|---|
| Test Case ID | TC-NFR-08-02 |
| Test Case Name | Non-finance role cannot use finance APIs |
| Related Requirement | NFR-08 (Role-Based Access Control), BR 10.1 |
| Related Test File | `tests/e2e/authorization.spec.ts` |
| Type | E2E (Security) |
| Priority | P1 – Must Pass |
| Preconditions | Logged in as SALES_MANAGER |
| Input | Call invoice and payment endpoints |
| Expected Output | 403 FORBIDDEN |
| Actual Output | Role-restriction test could not execute its assertions: E2E_INVOICE_ID was missing in the recorded run. |
| Status | Fail |

| Field | Content |
|---|---|
| Test Case ID | TC-NFR-08-03 |
| Test Case Name | Accountant cannot list users or leads |
| Related Requirement | NFR-08 (Role-Based Access Control), BR 10.1 |
| Related Test File | `tests/e2e/management-authorization.spec.ts` |
| Type | E2E (Security) |
| Priority | P1 – Must Pass |
| Preconditions | Logged in as ACCOUNTANT |
| Input | GET /users, GET /leads, POST /clients |
| Expected Output | 403 FORBIDDEN for each |
| Actual Output | Accountant-only checks skipped because E2E_ACCOUNT-ANT_EMAIL and E2E_ACCOUNTANT_PASSWORD were not configured. |
| Status | Skipped |